Alireza Fadaei

Full-Stack Software Engineer | React · FastAPI · Python · Applied AI

Oulu, Finland · Email Alireza · Portfolio · GitHub profile · LinkedIn · ORCID

PROFILE

Full-stack software engineer building practical products from frontend interfaces and backend services to data models, deployment and ongoing operation. Experience developing business applications, B2B SaaS platforms, research-data systems and security software, with recent work in React, FastAPI and AI-powered automation.

TECHNICAL SKILLS

Full-stack & APIs React · JavaScript · Python · FastAPI · Node.js · REST APIs · Django · Flask

Architecture & design Application architecture · Modular services · API design · System integration

Applied AI & automation Agentic AI workflows · LLM integration · AI-assisted development · Pydantic · Web crawling

Data & infrastructure PostgreSQL · SQL · Data modelling · Redis · Docker · Docker Compose · Linux · Nginx

Engineering practices Git · Code review · Integration testing · Technical documentation · Team collaboration

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer | Aminyx (Remote) · 2026

- Developed React interfaces and Python/FastAPI backend services for a modular B2B trade platform, connecting user workflows, REST APIs and business logic.
- Developed AI-assisted sourcing and trust-scoring workflows and a Dockerized company/contact-enrichment pipeline (Python, Pydantic, web crawling, optional LLM resolution and retryable checkpoints). Enrichment pipeline on GitHub

Full-Stack Software Developer | Fardaad (ISTT) · 2022–2024

- Built frontend and backend features for vulnerability- and patch-management applications in municipal and government networks, integrating user interfaces, APIs and operational data.
- Improved patch-deployment logic and backend algorithms; collaborated on application design, integration and Git-based code reviews.

Backend Developer & Systems Administrator | MetaMenu · 2021–2022

- Developed backend services and managed cloud-hosted infrastructure for a restaurant platform supporting augmented reality (AR) 3D menus, ordering, delivery and accounting workflows.

Full-Stack Research Software Developer | BrassicaLNC Project · 2021–2022

- Designed and developed frontend search and data-exploration views, backend services and research-data models; managed searchable records, dataset downloads and server operations for 1,852 curated lncRNAs from 2,361 RNA-Seq libraries (110 BioProjects). Explore BrassicaLNC

Backend Developer | Piranest Team · 2021–2022

- Built backend marketplace features and reservation workflows connecting travellers with local tour guides.

Python Web Developer | Fapsco Project · 2020

- Developed and maintained Flask applications and internal tools supporting client-facing services and business workflows.

SELECTED PROJECTS

ML-KEM Modernization — GitHub | University of Oulu course team project · 2026

- Designed the architecture and interfaces connecting a legacy device, edge gateway and cloud service; documented module boundaries and security/data flows.
- Coordinated five development workstreams using Docker Compose, Git and instructions for AI coding agents, with shared integration and testing expectations.

CNC Process Management System — live demo | Independent full-stack project

- Designed, built and deployed the frontend, backend, business logic and data models of a manufacturing operations system.
- Integrated process tracking, inventory visibility and workstation monitoring into end-to-end operational workflows.

Psychological Clinic CRM — live demo | Independent full-stack project

- Designed and built the frontend, backend and structured data model for appointments, patient records, clinician schedules, sessions and billing.
- Owned application architecture, frontend/backend integration and deployment for daily clinic operations.

TEACHING & KNOWLEDGE SHARING

Programming Instructor | Iran TVTO · 2024–2026

- Delivered hands-on programming and software-development instruction, sharing implementation and engineering practices.

EDUCATION

M.Sc. Software Engineering and Information Systems · University of Oulu, Finland · 2026–Present

M.Sc. Computer Science · Islamic Azad University, Najafabad Branch · 2022–2025

B.Sc. Computer Engineering · University of Isfahan · 2017–2022

RESEARCH PUBLICATION

Peer-reviewed paper (2026): Federated deep-learning intrusion detection for dew computing environments, The Journal of Supercomputing. Read publication (Springer)

RESEARCH INTERESTS

Federated & distributed learning · Intrusion detection & cybersecurity · Privacy-preserving machine learning · Reliable AI-enabled systems · Edge computing & IoT security · Software architecture & quality · Research software & scientific data platforms · Applied AI & LLM automation · Post-quantum security integration

LANGUAGES

English: professional proficiency (IELTS Academic 7.5, 2025) · Finnish: beginner (currently studying)